

Theories of Computation: Summative Assignment 1

To be handed in on Canvas before **Tuesday 7th March, 12pm**

Exercise 1 Let $\Sigma = \{a, b\}$ be the alphabet for this question. Let E be the regular expression aa^*bb^* .

1. Are the following words recognized by E ? [3 marks]

- (a) ϵ
- (b) aaa
- (c) $abbb$

2. Give a DFA with at most five states that accepts the language described by E . [3 marks]

NB. It can be a partial DFA.

3. Give a context-free grammar that generates the language described by E . [3 marks]

Exercise 2 Let $\Sigma = \{a, b\}$ be the alphabet for this question. Consider the following context-free grammar G :

$$\begin{array}{lcl} \Rightarrow S & ::= & aSa \mid B \\ B & ::= & BB \mid b \end{array}$$

1. Give a derivation tree for the word $abbba$ and the corresponding leftmost derivation in G . [3 marks]

2. What is the language $L(G)$ generated by this grammar? [3 marks]

3. Is $L(G)$ regular? Justify your answer. [3 marks]

Exercise 3 There are uncountably many colours in the ideal visible light spectrum. A computer science student had the idea to use the RGB gamut as a base to design a grammar to describe any colour as a string of: a number of R symbols, followed by a number of G symbols, followed by a number of B symbols. Will they succeed in describing all the colours? Justify your answer. [2 marks]